

1 Decompositions

$\mathbb{R}^2$, Lebesgue measures.

The joint probability density function

$$p(x_1, x_2) = \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1 - \rho^2)}} \exp \left[-\frac{1}{2} Q(x_1, x_2) \right]$$

where

$$Q(x_1, x_2) = \frac{\sigma_2^2 x_1^2 - 2\rho \sigma_1 \sigma_2 x_1 x_2 + \sigma_1^2 x_2^2}{\sigma_1^2 \sigma_2^2 (1 - \rho^2)}.$$

The marginal density function over X_1 is given by integrating over x_2 ,

$$\begin{aligned} p(x_1) &= \int dx_2 p(x_1, x_2) \\ &= \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1 - \rho^2)}} \int dx_2 \exp \left[-\frac{1}{2} Q(x_1, x_2) \right]. \end{aligned}$$

In order to evaluate this integral we need to isolate all of the x_2 terms in $Q(x_1, x_2)$. This requires completing the square for x_2 ,

$$\begin{aligned} \sigma_2^2 x_1^2 - 2\rho \sigma_1 \sigma_2 x_1 x_2 + \sigma_1^2 x_2^2 &= \sigma_1^2 \left(x_2 - \rho \frac{\sigma_2}{\sigma_1} x_1 \right)^2 + \sigma_2^2 x_1^2 - \rho \sigma_2^2 x_1^2 \\ &= \sigma_1^2 \left(x_2 - \rho \frac{\sigma_2}{\sigma_1} x_1 \right)^2 + \sigma_2^2 x_1^2 (1 - \rho^2). \end{aligned}$$

Then

$$\begin{aligned} Q(x_1, x_2) &= \frac{\sigma_2^2 x_1^2 - 2\rho \sigma_1 \sigma_2 x_1 x_2 + \sigma_1^2 x_2^2}{\sigma_1^2 \sigma_2^2 (1 - \rho^2)} \\ &= \frac{\sigma_1^2 \left(x_2 - \rho \frac{\sigma_2}{\sigma_1} x_1 \right)^2 + \sigma_2^2 x_1^2 (1 - \rho^2)}{\sigma_1^2 \sigma_2^2 (1 - \rho^2)} \\ &= \frac{\left(x_2 - \rho \frac{\sigma_2}{\sigma_1} x_1 \right)^2}{\sigma_2^2 (1 - \rho^2)} + \frac{x_1^2}{\sigma_1^2}. \end{aligned}$$

Consequently, we can write the joint density function as

$$\begin{aligned}
p(x_1, x_2) &= \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left[-\frac{1}{2}Q(x_1, x_2)\right] \\
&= \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left[-\frac{1}{2}\left(\frac{(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1)^2}{\sigma_2^2(1-\rho^2)} + \frac{x_1^2}{\sigma_1^2}\right)\right] \\
&= \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left[-\frac{1}{2}\frac{(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1)^2}{\sigma_2^2(1-\rho^2)}\right] \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right].
\end{aligned}$$

This factorization of the joint density function then allows us to reduce the marginal density function to a normal integral, which we can evaluate in closed form,

$$\begin{aligned}
p(x_1) &= \int dx_2 p(x_1, x_2) \\
&= \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \int dx_2 \exp\left[-\frac{1}{2}\frac{(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1)^2}{\sigma_2^2(1-\rho^2)}\right] \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right] \\
&= \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right] \int dx_2 \exp\left[-\frac{1}{2}\frac{(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1)^2}{\sigma_2^2(1-\rho^2)}\right] \\
&= \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right] \sqrt{2\pi\sigma_2^2(1-\rho^2)} \\
&= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right].
\end{aligned}$$

After all of the that, the marginal density function is just a normal density function,

$$\begin{aligned}
p(x_1) &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right] \\
&= \text{normal}(x_1 \mid 0, \sigma_1).
\end{aligned}$$

We can use this same factorization of the joint density function to simplify the calcu-

lation of the conditional density functions,

$$\begin{aligned}
p(x_2 | x_1) &= \frac{p(x_1, x_2)}{p(x_1)} \\
&= \frac{\frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left[-\frac{1}{2}\frac{(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1)^2}{\sigma_2^2(1-\rho^2)}\right] \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right]}{\frac{1}{\sqrt{2\pi}\sigma_1} \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right]} \\
&= \frac{1}{\sqrt{2\pi}\sigma_2^2(1-\rho^2)} \exp\left[-\frac{1}{2}\frac{(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1)^2}{\sigma_2^2(1-\rho^2)}\right].
\end{aligned}$$

Miraculously, each conditional density function is also another normal density function,

$$\begin{aligned}
p(x_2 | x_1) &= \frac{1}{\sqrt{2\pi}\sigma_2^2(1-\rho^2)} \exp\left[-\frac{1}{2}\frac{(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1)^2}{\sigma_2^2(1-\rho^2)}\right] \\
&= \text{normal}\left(x_2 \mid \rho\frac{\sigma_2}{\sigma_1}x_1, \sigma_2\sqrt{1-\rho^2}\right).
\end{aligned}$$

Because of the symmetric of the joint density function, the same calculations apply if when deriving the other marginal density function and corresponding conditional density functions. The result just swaps all of the indices,

$$\begin{aligned}
p(x_2) &= \text{normal}(x_2 \mid 0, \sigma_2) \\
p(x_1 | x_2) &= \text{normal}\left(x_1 \mid \rho\frac{\sigma_1}{\sigma_2}x_2, \sigma_1\sqrt{1-\rho^2}\right).
\end{aligned}$$

Finally, this is an example where some pattern matching can point to the marginal and conditional density functions directly. Examining

$$p(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left[-\frac{1}{2}\frac{(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1)^2}{\sigma_2^2(1-\rho^2)}\right] \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right],$$

we might notice that both of the exponentiated quadratics define normal density functions,

$$\begin{aligned}
p(x_1, x_2) &= \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp\left[-\frac{1}{2}\frac{\left(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1\right)^2}{\sigma_2^2(1-\rho^2)}\right] \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right] \\
&= \frac{1}{\sqrt{2\pi}\sigma_2^2(1-\rho^2)} \exp\left[-\frac{1}{2}\frac{\left(x_2 - \rho\frac{\sigma_2}{\sigma_1}x_1\right)^2}{\sigma_2^2(1-\rho^2)}\right] \frac{1}{\sqrt{2\pi}\sigma_1^2} \exp\left[-\frac{1}{2}\frac{x_1^2}{\sigma_1^2}\right] \\
&= \text{normal}\left(x_1 \mid \rho\frac{\sigma_1}{\sigma_2}x_2, \sigma_1\sqrt{1-\rho^2}\right) \text{normal}(x_2 \mid 0, \sigma_2).
\end{aligned}$$

In other words, by isolating x_2 we have incidentally manipulated the joint density function into the desired conditional decomposition. This approach isn't always useful, but it can save time when working with similar calculations.

2 Convolution 1

Start with $X = \mathbb{R}$ and probability distribution defined by the normal density function

$$\begin{aligned}
p(x) &= \text{normal}(x \mid \mu, \sigma) \\
&= \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left[-\frac{1}{2}\frac{(x - \mu)^2}{\sigma^2}\right].
\end{aligned}$$

We can lift this probability distribution into a joint distribution over X^2 by introducing a conditional probability kernel defined by the normal density functions

$$\begin{aligned}
p(x' \mid x) &= \text{normal}(x' \mid \eta - x, \tau) \\
&= \frac{1}{\sqrt{2\pi}\tau^2} \exp\left[-\frac{1}{2}\frac{(x' - \eta + x)^2}{\tau^2}\right].
\end{aligned}$$

The lifted joint density function is given by

$$\begin{aligned}
p(x, x') &= p(x' \mid x)p(x) \\
&= \text{normal}(x \mid \mu, \sigma) \text{normal}(x' \mid \eta - x, \tau) \\
&= \frac{1}{2\pi\sqrt{\sigma^2\tau^2}} \exp\left[-\frac{1}{2}Q(x, x')\right],
\end{aligned}$$

where

$$Q(x, x') = \frac{(x - \mu)^2}{\sigma^2} + \frac{(x' - \eta + x)^2}{\tau^2}.$$

The convolution of $p(x)$ with $p(x' | x)$ is given by projecting this joint density function to the auxiliary space,

$$\begin{aligned} p(x') &= \int dx p(x, x') \\ &= \frac{1}{2\pi\sqrt{\sigma^2\tau^2}} \int dx \exp\left[-\frac{1}{2}Q(x, x')\right]. \end{aligned}$$

To evaluate this integral, we'll need to isolate x with some completion of some squares,

$$\begin{aligned} Q(x, x') &= \frac{(x - \mu)^2}{\sigma^2} + \frac{(x' - \eta + x)^2}{\tau^2} \\ &= \frac{x^2 - 2\mu x + \mu^2}{\sigma^2} + \frac{x^2 - 2(x' - \eta)x + (x' - \eta)^2}{\tau^2} \\ &= \frac{1}{\sigma^2\tau^2} \left[(\sigma^2 + \tau^2)x^2 - 2(\tau^2\mu + \sigma^2(x' - \eta))x + \tau^2\mu^2 + \sigma^2(x' - \eta)^2 \right] \\ &= \frac{1}{\sigma^2\tau^2} \left[(\sigma^2 + \tau^2) \left(x - \frac{\tau^2\mu + \sigma^2(x' - \eta)}{\sigma^2 + \tau^2} \right)^2 \right. \\ &\quad \left. + \tau^2\mu^2 + \sigma^2(x' - \eta)^2 - \frac{(\tau^2\mu + \sigma^2(x' - \eta))^2}{\sigma^2 + \tau^2} \right] \\ &= \frac{1}{\sigma^2\tau^2} \left[(\sigma^2 + \tau^2) \left(x - \frac{\tau^2\mu + \sigma^2(x' - \eta)}{\sigma^2 + \tau^2} \right)^2 \right] \\ &\quad + \frac{1}{\sigma^2\tau^2} \left[\frac{\tau^4\mu^2 + \sigma^2\tau^2\mu^2 + \sigma^4(x' - \eta)^2 + \sigma^2\tau^2(x' - \eta)^2}{\sigma^2 + \tau^2} \right. \\ &\quad \left. - \frac{\tau^4\mu^2 + 2\tau^2\sigma^2\mu(x' - \eta) + \sigma^4(x' - \eta)^2}{\sigma^2 + \tau^2} \right] \\ &= \frac{1}{\sigma^2\tau^2} \left[(\sigma^2 + \tau^2) \left(x - \frac{\tau^2\mu + \sigma^2(x' - \eta)}{\sigma^2 + \tau^2} \right)^2 \right] \\ &\quad + \frac{1}{\sigma^2\tau^2} \left[\frac{\sigma^2\tau^2\mu^2 - 2\tau^2\sigma^2\mu(x' - \eta) + \sigma^2\tau^2(x' - \eta)^2}{\sigma^2 + \tau^2} \right] \\ &= \frac{\sigma^2 + \tau^2}{\sigma^2\tau^2} \left(x - \frac{\tau^2\mu + \sigma^2(x' - \eta)}{\sigma^2 + \tau^2} \right)^2 \\ &\quad + \frac{\mu^2 - 2\mu(x' - \eta) + (x' - \eta)^2}{\sigma^2 + \tau^2} \\ &= \frac{\sigma^2 + \tau^2}{\sigma^2\tau^2} \left(x - \frac{\tau^2\mu + \sigma^2(x' - \eta)}{\sigma^2 + \tau^2} \right)^2 \\ &\quad + \frac{(x' - (\mu + \eta))^2}{\sigma^2 + \tau^2}. \end{aligned}$$

Now we can reduce the convolution operation to a normal integral,

$$\begin{aligned}
p(x') &= \frac{1}{2\pi\sqrt{\sigma^2\tau^2}} \int dx \exp\left[-\frac{1}{2}Q(x, x')\right] \\
&= \frac{1}{2\pi\sqrt{\sigma^2\tau^2}} \int dx \exp\left[-\frac{1}{2}\frac{\sigma^2+\tau^2}{\sigma^2\tau^2}\left(x - \frac{\tau^2\mu + \sigma^2(x' - \eta)}{\sigma^2 + \tau^2}\right)^2\right] \exp\left[-\frac{1}{2}\frac{(x' - (\mu + \eta))^2}{\sigma^2 + \tau^2}\right] \\
&= \frac{1}{2\pi\sqrt{\sigma^2\tau^2}} \exp\left[-\frac{1}{2}\frac{(x' - (\mu + \eta))^2}{\sigma^2 + \tau^2}\right] \int dx \exp\left[-\frac{1}{2}\frac{\sigma^2+\tau^2}{\sigma^2\tau^2}\left(x - \frac{\tau^2\mu + \sigma^2(x' - \eta)}{\sigma^2 + \tau^2}\right)^2\right] \\
&= \frac{1}{2\pi\sqrt{\sigma^2\tau^2}} \exp\left[-\frac{1}{2}\frac{(x' - (\mu + \eta))^2}{\sigma^2 + \tau^2}\right] \sqrt{2\pi\frac{\sigma^2\tau^2}{\sigma^2 + \tau^2}} \\
&= \frac{1}{\sqrt{2\pi(\sigma^2 + \tau^2)}} \exp\left[-\frac{1}{2}\frac{(x' - (\mu + \eta))^2}{\sigma^2 + \tau^2}\right] \\
&= \text{normal}\left(x' \mid \mu + \eta, \sqrt{\sigma^2 + \tau^2}\right).
\end{aligned}$$

This has a neat interpretation. The convolution operation takes in initial normal density function, translates it by η , and then inflates the standard deviation.

In the limit $\tau \rightarrow 0$, the convolution results in a pure translation,

$$p(x') = \text{normal}\left(x' \mid \mu + \eta, \sqrt{\sigma^2 + \tau^2}\right).$$

Specifically, this is exactly the pushforward of a normal density function along the translation operation that we discussed in **Section Blah**.

At the same time, the convolution kernel in this limit acts like a singular Delta function that shifts an input x to $x + \eta$. This limit is also known as a linear convolution.

3 Convolution 2

Start with $X = \mathbb{I}$ and

$$p(x) = \text{Poisson}(x \mid \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

We can left this probability distribution into a joint distribution over X^2 by introducing a conditional probability kernel defined by the conditional density functions

$$p(x' \mid x) = \frac{\delta^{x'-x} e^{-\delta}}{(x' - x)!} I[x \leq x'].$$

These conditional density functions look a bit awkward. Let's verify that they're prop-

erly normalized for all δ ,

$$\begin{aligned}
\sum_{x'=0}^{\infty} p(x' | x) &= \sum_{x'=0}^{\infty} \frac{\delta^{x'-x} e^{-\delta}}{(x' - x)!} I[x \leq x'] \\
&= \sum_{x'=x}^{\infty} \frac{\delta^{x'-x} e^{-\delta}}{(x' - x)!} \\
&= \sum_{k=0}^{\infty} \frac{\delta^k e^{-\delta}}{k!} \\
&= e^{-\delta} \sum_{k=0}^{\infty} \frac{\delta^k}{k!} \\
&= e^{-\delta} e^{\delta} \\
&= 1.
\end{aligned}$$

With the validity of the convolution kernel verified, we can proceed to the convolution itself,

$$\begin{aligned}
p(x') &= \sum_{x=0}^{\infty} p(x' | x) p(x) \\
&= \sum_{x=0}^{\infty} \frac{\delta^{x'-x} e^{-\delta}}{(x' - x)!} I[x \leq x'] \frac{\lambda^x e^{-\lambda}}{x!} \\
&= e^{-(\lambda+\delta)} \sum_{x=0}^{\infty} I[x \leq x'] \frac{1}{x! (x' - x)!} \delta^{x'-x} \lambda^x \\
&= \frac{e^{-(\lambda+\delta)}}{x'!} \sum_{x=0}^{x'} \frac{x'!}{x! (x' - x)!} \delta^{x'-x} \lambda^x.
\end{aligned}$$

This summation can be simplified by the binomial theorem,

$$(a + b)^K = \sum_{k=0}^K a^k b^{K-k}.$$

Consequently,

$$\begin{aligned}
p(x') &= \frac{e^{-(\lambda+\delta)}}{x'!} \sum_{x=0}^{x'} \frac{x'!}{x! (x' - x)!} \delta^{x'-x} \lambda^x \\
&= \frac{e^{-(\lambda+\delta)}}{x'!} (\lambda + \delta)^{x'}.
\end{aligned}$$

This, however, is just another Poisson density function,

$$p(x') = \frac{e^{-(\lambda+\delta)}}{x'!}(\lambda + \delta)^{x'} = \text{Poisson}(x' \mid \lambda + \delta).$$

This particular convolution translates the intensity of the Poisson parameter.